

Math 421

Friday, November 21

Announcements

- Homework 9 due **today**
- Next week's drop-in hours:
 - **Tuesday** 9-11 am VV B205
 - Wednesday 12-1 pm VV 311
- Check for any final exam conflicts & email me

Chapter 3: Continuity

Section 19: Uniform Continuity

Uniform Continuity

Definition. Let f be a real-valued function defined on a set $S \subseteq \mathbb{R}$. Then f is **uniformly continuous** on S if:

for all $\epsilon > 0$ there exists $\delta > 0$ such that

$$\boxed{x, y \in S} \text{ and } \underbrace{|x - y| < \delta} \text{ imply } \underbrace{|f(x) - f(y)| < \epsilon}.$$

Q: If f is uniformly continuous on its domain, is f continuous on its domain?

uniformly continuous seems stricter.

Yes.

Set $y = x_0$

Uniform Continuity Proof Recipe

To show that $f: S \rightarrow \mathbb{R}$ is uniformly continuous on S :

- I. “Fix $\epsilon > 0$.”
- II. “Let $\delta = (\textit{something})$.” [this requires scratch work]
- III. “Assume $|x - y| < \delta$ and $x, y \in S$.”
- IV. Show that $|f(x) - f(y)| < \epsilon$.
- V. “Since $\epsilon > 0$ was arbitrary, f is uniformly continuous on S .”

Example

To prove cont. of x^2 on $\mathbb{R}$:

$$\delta = \min \left\{ 1, \frac{\varepsilon}{2|x_0|+1} \right\}$$

↳ depends on ε and x_0 .

Let $S = \{x \in \mathbb{R} : -4 \leq x \leq 4\}$, show that $g(x) = x^2$ is uniformly continuous on S .

Discussion: $x, y \in [-4, 4]$

$$|x-y| < \delta = \frac{\varepsilon}{8}$$

↳ should only depend on ε .

$$|x^2 - y^2| = |x-y||x+y| < \varepsilon$$
$$< \delta \cdot 8$$

$$|x+y| \leq 8$$

Proof: Let $\varepsilon > 0$. Set $\delta = \frac{\varepsilon}{8}$. Assume $|x-y| < \delta$ and $x, y \in [-4, 4]$.

$$\text{Then } |x^2 - y^2| = |x-y||x+y| < \delta \cdot 8 = \frac{\varepsilon}{8} \cdot 8 = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $g(x) = x^2$ is uniformly continuous on $[-4, 4]$.

Activity – Find δ

1. Show that any constant function is uniformly continuous on $\mathbb{R}$.

$\delta = \varepsilon, 1, \frac{\varepsilon}{|c|}$ "let δ be any number $\delta > 0$ ".

2. Let $S = \{x \in \mathbb{R} : -N \leq x \leq N\}$, show that $g(x) = x^2$ is uniformly continuous on S .

$$\delta = \frac{\varepsilon}{2N}$$

Q: Do you think $g(x) = x^2$ is uniformly continuous on $\mathbb{R}$?

No!

Continuity and Uniform Continuity

Fact. If f is uniformly continuous on its domain, then it is continuous on its domain.

Theorem 19.2. If f is a continuous function on $[a, b]$, then f is uniformly continuous on $[a, b]$.

Chapter 5: Differentiation

Section 28: Basic Properties of the Derivative

Differentiable

Definition. Let f be a real-valued function defined on an open interval containing a point a . We say f is **differentiable at a** , or f has a derivative at a , if the limit

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

exists and is finite. $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$. domain of f' is $\{a \in \text{dom } f : f'(a) \text{ exists}\}$

Example

Compute f' if $f(x) = x^2$.

Let $a \in \mathbb{R}$.

$$\begin{aligned} \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} &= \lim_{x \rightarrow a} \frac{x^2 - a^2}{x - a} = \lim_{x \rightarrow a} \frac{(x + a)(x - a)}{x - a} \\ &\stackrel{*}{=} \lim_{x \rightarrow a} (x + a) = 2a. \end{aligned}$$

So $f'(a) = 2a$ for all $a \in \mathbb{R}$.

* this equality only holds because the limit does not care about what happens at a itself. The function $\frac{(x + a)(x - a)}{x - a}$ is not equal to $(x + a)$.

• In ch. 5 and beyond, can use ^{algebraic} limit laws/properties/theorems without comment.

Activity

Compute f' if $f(x) = mx + b$, where m, b are constants.

Note: If x is close to a if $a > 0, x > 0$
if $a < 0, x < 0$.

Activity – Divide & Conquer

Let $f(x) = |x|$. Compute f' by dividing into 3 cases.

Case 1: $a > 0$

Case 2: $a < 0$

Case 3: $a = 0$

